

Xinghao Chen

Ph.D. Student in Computer Science

Research Interests:

Thought Compression & Distillation

Latent-Space Reasoning

Efficient Memory-Augmented Agents



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🏠 [Google Scholar](#)

EDUCATION

The Hong Kong Polytechnic University & Eastern Institute of Technology, Ningbo

Joint Ph.D. Student in Computer Science

Hong Kong / Ningbo, China

Sep. 2024 – Present

Supervisors: [Wenjie Li](#), Professor [Xiaoyu Shen](#), Assistant Professor

Nankai University, College of Artificial Intelligence

B.S. in Intelligent Science and Technology

Tianjin, China

Sep. 2019 – Jun. 2023

RESEARCH EXPERIENCE

Tencent Youtu Lab

Research Intern

Jun. 2026 – Present

Working with: [Junnan Dong](#)

Eastern Institute for Advanced Study, College of Information Science and Technology

Research Assistant

Ningbo, China

Jan. 2024 – Aug. 2024

Supervisor: [Xiaoyu Shen](#), Assistant Professor

The Hong Kong Polytechnic University

Research Assistant

Hong Kong, China

Sep. 2023 – Jan. 2024

Supervisor: [Xinyan Huang](#), Associate Professor

PUBLICATIONS

* indicates equal contribution, † indicates corresponding author.

LLM Reasoning & Efficient Intelligence

- [1] [X. Chen](#), C.T. Leong, W. Guo, J. Wang, W. Li, X. Shen[†]. What Makes Effective Supervision in Latent Chain-of-Thought: An Information-Theoretic Analysis. [C]// (ICML) Forty-Third International Conference on Machine Learning. **CCF-A**
Established an information-theoretic framework dissecting latent CoT supervision into Trajectory Control and State Alignment; introduced the Unified Latent Probe (ULP) to reveal a strict information-performance binding between mutual information retention and reasoning accuracy.
- [2] [X. Chen](#)^{*}, A. Zhao^{*}, H. Xia, X. Lu, H. Wang, Y. Chen, W. Zhang, J. Wang[†], W. Li, X. Shen[†]. Reasoning Beyond Language: A Comprehensive Survey on Latent Chain-of-Thought Reasoning. [C]// (EMNLP) Findings of the 2026 Conference on Empirical Methods in Natural Language Processing. **CCF-B**
The first comprehensive survey on latent CoT reasoning, categorizing methods along two axes: *Token-wise Horizontal Level* and *Layer-wise Vertical Level*; maintaining a continuously updated repository at [Awesome-Latent-CoT](#).
- [3] [X. Chen](#), Z. Sun, W. Guo, M. Zhang, Y. Chen, Y. Sun, H. Su, Y. Pan, D. Klakow, W. Li[†], X. Shen[†]. Unveiling the Key Factors for Distilling Chain-of-Thought Reasoning. [C]// (ACL) Findings of the 2025 Annual Meeting of the Association for Computational Linguistics. **CCF-A**
Conducted systematic study on *what makes effective Chain-of-Thought supervision for smaller models*, revealing that the conventional wisdom of “stronger teacher, better student” does not hold in reasoning distillation.
- [4] S. Lyu^{*}, [X. Chen](#)^{*}, Z. Sun, T. Liu, D. Zhu, X. Shen. When Compression Helps and When It Hurts: Condition-Aware Analysis of Chain-of-Thought Distillation. [C]// (EMNLP) The 2026 Conference on Empirical Methods in Natural Language Processing. **CCF-B**
- [5] H. Chen^{*}, J. Lin^{*}, [X. Chen](#), Y. Fan, X. Jin, H. Su, J. Dong, J. Fu, X. Shen. Rethinking Visual Layer Selection in Multimodal LLMs. [C]// (EMNLP) The 2025 Conference on Empirical Methods in Natural Language Processing. **CCF-B, Oral**
- [6] Y. Chen, D. Zhu, Y. Sun, [X. Chen](#), W. Zhang, X. Shen. The Accuracy Paradox in RLHF: When Better Reward Models Don't Yield Better Language Models. [C]// (EMNLP) The 2024 Conference on Empirical Methods in Natural Language Processing. **CCF-B**
- [7] X. Lu, S. Liu, B. Yin, Y. Li, [X. Chen](#), H. Su, Y. Jin, W. Zeng, X. Shen. MultiConIR: Towards Multi-Condition Information Retrieval. [C]// (EMNLP) Findings of the 2025 Conference on Empirical Methods in Natural Language Processing. **CCF-B**
- [8] H. Wu, X. Wang, J. Zhang, J. Tong, [X. Chen](#), J. Lin, Y. Ma, X. Shen. UTPTrack: Towards Simple and Unified Token Pruning for Visual Tracking. [C]// (CVPR) The 2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition. **CCF-A**
- [9] J. Yang, [X. Chen](#), Y. Liu, Z. Sun, W. Guo, X. Shen. AutoDataset: A Lightweight System for Continuous Dataset Discovery and Search. Arxiv, 2603.07271.

Applied AI (Pre-doctoral)

- [10] X. Zhang, [X. Chen](#), Y. Ding, Y. Zhang, Z. Wang, J. Shi, N. Johansson, X. Huang. [Smart Real-Time Evaluation of Tunnel Fire Risk and Evacuation Safety via Computer Vision](#). *Safety Science*, 2024. **CAS Q1, JCR Q1**
- [11] Y. Ding, [X. Chen](#), Y. Zhang, X. Huang. [Smart Building Evacuation by Tracking Multi-Camera Network and Explainable Re-Identification Model](#). *Engineering Applications of Artificial Intelligence*, 2025. **CAS Q1, JCR Q1**
- [12] Y. Ding, [X. Chen](#), Z. Wang, Y. Zhang, X. Huang. [Human Behaviour Detection Dataset \(HBDset\) Using Computer Vision for Evacuation Safety and Emergency Management](#). *Journal of Safety Science and Resilience*, 2024.
- [13] [X. Chen](#), G. Cheng, S. Liu, S. Meng, Y. Jiao, W. Zhang, J. Liang, W. Zhang. Probing 1D Convolutional Neural Network Adapted to Near-Infrared Spectroscopy for Efficient Classification of Mixed Fish. *Spectrochimica Acta Part A*, 2022. **CAS Q2, JCR Q1**

RESEARCH PROJECTS

Latent Chain-of-Thought Reasoning

Jan. 2025 – Feb. 2026

Core research direction during Ph.D.

Investigating how to internalize explicit reasoning into continuous latent spaces for efficient and faithful LLM inference. Established an information-theoretic framework for analyzing supervision signals in latent CoT [1]; authored the first comprehensive survey of the field [2]; and empirically identified key factors governing CoT distillation from large to small models [3]. Ongoing work explores extending latent reasoning representations to serve as compressed, reusable memory for agentic systems.

Efficient Latent Memory for Long-Horizon Tool-Use Agents

Feb. 2026 – Present

Ongoing research

Current tool-use agents suffer from two compounding memory bottlenecks: *intra-session*, where KV caches grow linearly and existing compression methods lose relational structure or remain session-bound; and *inter-session*, where successful trajectories are verbalized into lossy text summaries that discard the activation-level structure of effective tool selections. We propose a unified latent memory framework that simultaneously compresses working memory, preserves relational structure, and remains semantically stable across session boundaries.

SERVICE

Teaching Assistant, The Hong Kong Polytechnic University

- COMP 3122: Information Systems Development Spring 2026
- COMP 5241: Software Engineering and Development Fall 2025
- COMP 1411: Introduction to Computer Systems Spring 2025
- COMP 5511: Artificial Intelligence Concepts Fall 2024

INVITED TALKS

Reasoning Beyond Language: A Comprehensive Survey on Latent CoT Reasoning

Invited talk at [NICE](#) and [CIPS](#) [[Video](#)]

Jun. 2025

SKILLS

- **Programming Languages:** Python, MATLAB, C++
- **LLM Frameworks:** LLaMA-Factory, VERL, vLLM, SGLang
- **AI-Assisted Development:** Claude Code, Cursor, OpenAI Codex, OpenClaw
- **Languages:** Mandarin (Native), English (Fluent)

INTERESTS & HOBBIES

- Drum Kit, Anime, Films

SELF-EVALUATION

- Endlessly curious about new ideas and technologies. Committed to research that matters beyond the paper — bridging the gap from insight to real-world impact. Collaborative, outgoing, and motivated by a simple standard: work that is technically sound, practically grounded, and genuinely worth sharing.